

MODULE 31

Kafka Integration with Spring Boot

Learn how to build scalable, event-driven applications by integrating Apache Kafka with Spring Boot -- producers, consumers, message handling, error management, and best practices for reliable event processing.





WHY SPRING KAFKA MATTERS

Spring Kafka simplifies integration with Apache Kafka and empowers developers to build scalable, reliable, and maintainable event-driven applications.

WHAT SPRING KAFKA GIVES YOU

- **Seamless Integration** — native support and abstractions that make it easy to integrate Kafka with Spring Boot.
- **Developer Productivity** — high-level APIs like `KafkaTemplate` and `@KafkaListener` reduce boilerplate code.
- **Reliability & Robustness** — built-in features for error handling, retries, dead letter queues, and offset management.
- **Scalability** — easily build applications that scale horizontally to handle millions of events.
- **Enterprise Ready** — production-grade features, observability support, and strong community adoption.

BUSINESS IMPACT

- **Real-Time Responsiveness** — process events instantly and respond to business changes as they happen.
- **Decoupled Applications** — reduce dependencies between services and increase flexibility.
- **Improved Reliability** — ensure message delivery and consistency with robust error handling.
- **Higher Throughput** — handle large volumes of events efficiently without impacting performance.
- **Lower Costs** — optimize resource usage and reduce operational overhead.

SPRING KAFKA HELPS YOU FOCUS ON BUSINESS LOGIC

Simple APIs

Out-of-the-Box Configuration

Flexible & Extensible

Monitoring & Observability

Large Community & Ecosystem

KEY TAKEAWAY Spring Kafka bridges the power of Apache Kafka with the productivity of the Spring ecosystem -- helping you build modern, event-driven applications faster and with confidence.

MODULE LEARNING OBJECTIVES

By the end of this module, you will be able to design, build, and validate a complete Kafka integration in a Spring Boot application.



1. Spring Kafka Fundamentals

Explain the role of Spring Kafka and its key components.



2. Integration Architecture

Understand message flow between producers, cluster, and consumers.



3. Kafka Producers

Build producers using KafkaTemplate with proper serialization.



4. Kafka Consumers

Implement consumers with @KafkaListener and consumer groups.



5. Advanced Messaging

Use message keys, offsets, error handling, and DLQs.



6. Reliable Processing

Achieve idempotency and at-least-once / exactly-once delivery.



7. Monitor and Optimize

Analyze performance and tune configurations for throughput.



8. Best Practices

Apply enterprise practices for topics, security, and scalability.



9. End-to-End Integration

Build a complete app that publishes and consumes events.



10. Hands-On Lab

Build, test, and validate Kafka integration in the graded lab.

MODULE DURATION

Theory: 1 Hour

Lab: 2.5 Hours

Total: 3.5 Hours

KEY TAKEAWAY You will be equipped with the skills to seamlessly integrate Kafka with Spring Boot and build scalable, reliable, and event-driven applications ready for production.

BUSINESS SCENARIO: EVENT-DRIVEN CUSTOMER NOTIFICATIONS

Northstar CRM must notify downstream systems the moment a customer is created or changes status -- without losing events or double-processing replays.

THE PROBLEM

- Agents create and update customers through the REST API built in Modules 24-29 -- but notification and audit systems learn nothing unless they poll the database.
- Leadership now requires asynchronous notification: creating Amina or activating Ravi must trigger downstream processes in real time, not on a schedule.
- Duplicate delivery (rebalance, retry, replay) must never send duplicate SMS/email side effects, and an invalid contract must never retry forever.

THE FIX: A VERSIONED CUSTOMEREVENT

- Publish a keyed CustomerEvent v1 after every successful customer write -- the event reflects a fact that already happened.
- Consume idempotently -- process each unique eventId exactly once for business-effect purposes, even if Kafka redelivers it.
- Recover poison messages (bad key, unsupported version) to a Dead Letter Topic after a bounded number of retries.

FIXTURES USED THROUGHOUT THIS MODULE

ID / Key	Value	Role	Notes
CUS-1001	Amina Khan	Primary publish/consume fixture	Status ACTIVE
CUS-1002	Ravi Singh	Second key / status-update fixture	Status PROSPECT
crm.customer-events.v1	Kafka topic	Primary event topic (from Lab 30)	Keyed by customerId
lab-request-001	correlationId	Traceability on every event and log line	Also used in Module 29

KEY TAKEAWAY This module builds the exact event backbone graded in Lab 31: KafkaTemplate publish, @KafkaListener consume, idempotent processing, and DLT recovery for Northstar CRM.



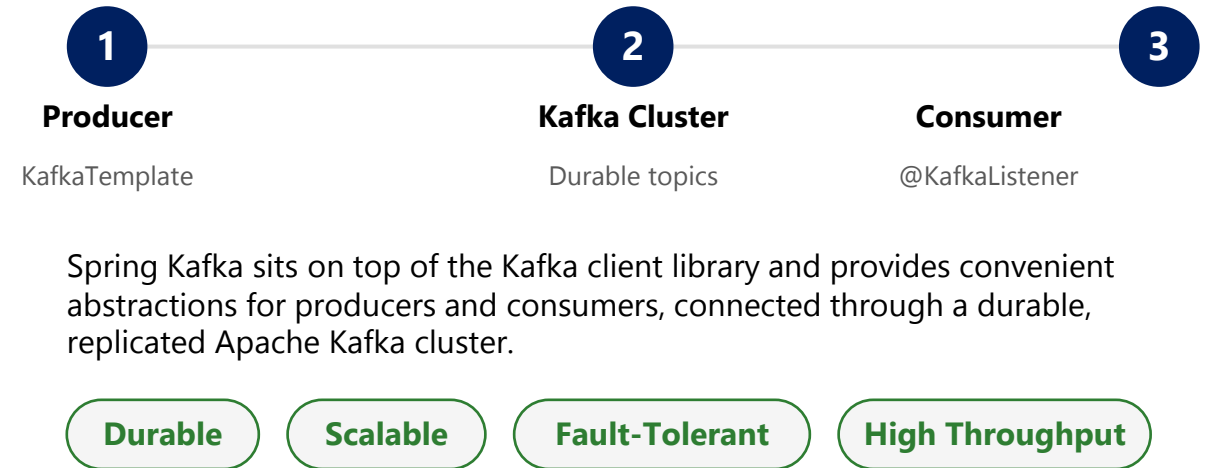
INTRODUCTION TO SPRING KAFKA

Spring Kafka is the official Spring project for building applications that use Apache Kafka for messaging and stream processing.

WHAT IS SPRING KAFKA?

- **Simplifies Development** — a Spring project that simplifies building Kafka-based messaging solutions.
- **High-Level Abstractions** — sits over the native Kafka client, reducing boilerplate code.
- **Deep Spring Integration** — integrates with Spring Boot, Spring Framework, and Spring Cloud.
- **Production Ready** — makes it easy to build scalable, reliable, event-driven applications.

HOW SPRING KAFKA WORKS



WHY IT MATTERS



KEY TAKEAWAY Spring Kafka provides developer-friendly APIs and integrations that make working with Apache Kafka in Spring applications productive and efficient.

KAFKA INTEGRATION ARCHITECTURE

Spring Boot applications use Spring Kafka to produce and consume events from Apache Kafka, enabling scalable, reliable, and decoupled systems.

KEY COMPONENTS

- **Producers** — Spring Boot services publish events using `KafkaTemplate`.
- **Kafka Cluster** — a distributed, fault-tolerant platform that stores and streams events.
- **Consumers** — Spring Boot services consume events using `@KafkaListener`.
- **Serialization** — events are serialized (JSON, Avro) for interoperability.
- **External Systems** — databases, caches, or services that interact with producers/consumers.

END-TO-END FLOW

1. Producer service creates an event.
2. Event is published to a Kafka topic.
3. Kafka brokers store and replicate the event.
4. Consumer service(s) read the event.
5. Event is processed and action is taken.

BENEFITS

- **Decoupled** — services communicate asynchronously via events.
- **Scalable** — producers and consumers scale independently.
- **Reliable** — durable storage and replication ensure no data loss.
- **High Throughput** — Kafka handles millions of events per second.

In Northstar CRM: the `CustomerController/Service` persists the customer, then `CustomerEventPublisher` (`KafkaTemplate`) publishes to `crm.customer-events.v1` -- consumed by `CustomerEventListener`.

KEY TAKEAWAY Spring Boot services publish and consume events through Kafka, enabling scalable, reliable, and decoupled systems across the enterprise application landscape.

SPRING KAFKA COMPONENTS

Spring Kafka provides a set of core components and abstractions that simplify building reliable and scalable Kafka-based applications.

CORE COMPONENTS

Component	Role
KafkaTemplate	High-level abstraction for producing messages to Kafka topics.
@KafkaListener	Annotation-based programming model for consuming messages.
Consumer Group	Multiple consumers working together to consume messages from topics.
ConsumerFactory / ProducerFactory	Factories that create configured Kafka producer and consumer instances.
Serialization / Deserialization	Converts objects to/from bytes using Key/Value Serializer and Deserializer.
KafkaProperties	Holds Kafka configuration properties from application.properties/yml.
MessageConverter	Converts messages between Java objects and Kafka record payloads.
Error Handling Components	Provides error handling, retries, and Dead Letter Queue (DLQ) support.

SUPPORTING COMPONENTS

Ack Mode

Offset Management

Message Key

RecordInterceptor

Metrics & Observation

KEY TAKEAWAY Spring Kafka components provide high-level abstractions, reduce boilerplate code, and help you build robust, scalable, and maintainable event-driven applications with ease.

TRY IT YOURSELF — EXERCISE 1: SPRING KAFKA ROLES

Reinforces: mapping Kafka concepts to Spring Boot components -- KafkaTemplate, @KafkaListener, bootstrap-servers, group-id.

STEP-BY-STEP

STEP	WHAT TO DO
Goal	Connect KafkaTemplate and @KafkaListener to producer/consumer concepts.
Step 1	Study the reference mapping (right) and copy it into notes/lab31-spring-kafka.md.
Step 2	CRM story: after HTTP creates Amina, the service calls KafkaTemplate to publish to crm.customer-events.v1 keyed by CUS-1001.
Step 3	Listener story: the notifications listener uses group crm-notifications and processes the JSON envelope.
Step 4	Gap check: list one open question about serializers (String vs. JSON) before lab.
Deliverable	Role-mapping notes + CRM produce/consume story + one serializer question (table copied, both stories written, question listed).

REFERENCE: KAFKA IDEA -> SPRING PIECE

```
// Kafka idea -> Spring Boot piece
Produce record    -> KafkaTemplate.send(...)
Consume record    -> @KafkaListener
Bootstrap servers -> spring.kafka.bootstrap-servers
Group id          -> spring.kafka.consumer.group-id

// CRM story (notes/lab31-spring-kafka.md)
produce: HTTP creates Amina -> KafkaTemplate
      .send("crm.customer-events.v1",
            "CUS-1001", event)
listen: group=crm-notifications processes
       the JSON envelope
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 1 before continuing: exercises/exercise-01-spring-kafka-roles.md.

KAFKA PRODUCER ARCHITECTURE

A Kafka producer application publishes events to topics in a Kafka cluster. Spring Kafka simplifies producer development and configuration.

PRODUCER FLOW

- **1. Create Message** — the application creates a domain event or data payload.
- **2. Send via KafkaTemplate** — KafkaTemplate sends the message with optional key, headers, and partition.
- **3. Serialization** — KeySerializer and ValueSerializer convert objects to bytes.
- **4. Partitioner** — the message is assigned to a partition based on key or strategy.
- **5. Broker Acknowledgement** — the message is sent to the leader broker and acknowledged per the acks policy.

PRODUCER CONFIGURATION (application.yml)

```
spring:
  kafka:
    bootstrap-servers: ${KAFKA_BOOTSTRAP_SERVERS:localhost:9092}
    producer:
      properties:
        enable.idempotence: true
        acks: all
        retries: 3
```

acks	Description	Durability
0	No acknowledgement	Low
1	Leader acknowledges	Medium
all (-1)	All in-sync replicas acknowledge	High

KEY TAKEAWAY Spring Kafka and KafkaTemplate work together to efficiently publish messages to Kafka topics with serialization, batching, partitioning, and fault-tolerant delivery.



KAFKATEMPLATE OVERVIEW

KafkaTemplate is the central class in Spring Kafka for sending messages. It provides a simplified, thread-safe API over the native Kafka Producer.

KEY CHARACTERISTICS

- Thread-safe -- one KafkaTemplate bean can be shared across the whole application.
- Uses configured serializers, interceptors, and producer properties automatically.
- Supports both synchronous (blocking) and asynchronous (non-blocking) sends.
- Returns a CompletableFuture carrying send metadata (partition, offset) for async handling.
- Easily configured through standard Spring Boot application properties.

LAB 31: CustomerEventPublisher.java

```
@Service
public class CustomerEventPublisher {
    private final KafkaTemplate<String, CustomerEvent> template;

    public void publish(String topic, CustomerEvent event) {
        template.send(topic, event.customerId(), event)
            .whenComplete((result, error) -> {
                if (error != null)
                    log.error("customer_event_publish_failed id={}",
                        event.eventId(), error);
                else
                    log.info("customer_event_published id={} offset={}",
                        event.eventId(), result.getRecordMetadata().offset());
            });
    }
}
```

KEY TAKEAWAY KafkaTemplate is the recommended way to publish messages in Spring Kafka. It abstracts the complexity of the native Kafka Producer while providing reliable, efficient event publishing.

PUBLISHING MESSAGES

Publishing messages means sending data from a Spring Boot application to a Kafka topic so that other services can consume and act on it.

KEY POINTS

- Use `KafkaTemplate` to send messages -- never the raw `KafkaProducer` client directly.
- Messages are published to a named topic (`crm.customer-events.v1` in Lab 31).
- Kafka handles durability, replication, and ordering within a partition.
- Publishing after a successful database write reflects a fact that already happened.
- Always handle the send result (success/failure); never publish and forget.

COMMON SEND OPTIONS

Option	Purpose
Topic	The destination topic where the message is published.
Key	Used for partitioning and ordering -- <code>customerId</code> in Lab 31.
Value (payload)	The actual message content -- the <code>CustomerEvent</code> .
Headers	Metadata about the message (e.g. <code>correlationId</code> , <code>eventType</code>).
Partition	Let Kafka decide by key hash (default), or specify one explicitly.

Wire the publisher into the customer create/status-change service methods so events are published only after the CRM database write succeeds -- never speculatively before it.

Use meaningful keys

Keep payloads small

Handle send results

Prefer async sends

KEY TAKEAWAY Publishing messages moves a fact from your application into Kafka's durable, replicated log so that other services can consume and act on it independently.

SERIALIZING EVENTS

Serialization converts Java objects into a byte array so they can be transmitted over the network and stored in Kafka.

WHY SERIALIZATION MATTERS

- **Network Communication** — Kafka brokers store and transfer data as bytes.
- **Interoperability** — different services and languages can read the same data.
- **Efficiency** — compact formats reduce payload size and network overhead.
- **Schema Evolution** — controlled changes to the event shape without breaking consumers.

Serializer	Use Case
StringSerializer	For simple text messages / message keys.
JsonSerializer	For JSON payloads -- recommended for microservices (Lab 31).
ByteArraySerializer	For raw byte[] messages.

SPRING KAFKA CONFIGURATION (Lab 31)

```
spring:
  kafka:
    consumer:
      group-id: crm-notifications
      auto-offset-reset: earliest
      enable-auto-commit: false
      properties:
        spring.json.trusted.packages: com.northstar.crm.event
    producer:
      properties:
        enable.idempotence: true
```

- Restrict trusted packages
- Keep contracts stable
- Version events (v1)
- Never expose internals

KEY TAKEAWAY Serialization is the bridge between your application objects and Kafka's byte-based messaging system -- choose the right serializer, keep contracts stable, and publish meaningful, versioned events.

KAFKA CONSUMER ARCHITECTURE

A Kafka consumer application subscribes to topics, processes messages, and commits offsets to ensure reliable, at-least-once (or exactly-once) processing.

CONSUMER FLOW



KEY CONSUMER CONCEPTS

- **Consumer Group** — enables multiple instances to share the work (crm-notifications in Lab 31).
- **Partitions** — messages in a topic are split across partitions for parallelism.
- **Offset** — a unique, monotonically increasing position within a partition.
- **At-Least-Once** — the default delivery guarantee; exactly-once needs idempotent design.

OFFSET COMMIT STRATEGIES

Strategy	Description
Auto Commit	Offsets committed automatically at intervals (default).
Manual Commit	Application commits after successful processing (Lab 31).
Commit Sync	Blocks until the broker acknowledges the commit.
Commit Async	Commits in the background (non-blocking).

Lab 31 consumer config

```
group-id: crm-notifications
enable-auto-commit: false
auto-offset-reset: earliest
```

Scalable

Reliable

Flexible

Resilient

KEY TAKEAWAY Kafka consumers enable your applications to reliably subscribe, process, and track progress of messages. Proper offset management ensures data is not lost and processing is reliable.

@KAFKALISTENER ANNOTATION

@KafkaListener is the core annotation in Spring Kafka used to declare a method that listens to one or more topics and processes incoming messages.

KEY FEATURES

- Simplifies consumer configuration -- Spring creates the consumer container for you.
- Supports topic subscription, consumer group, partitions, and concurrency.
- Supports single record, batch, or manual-acknowledgment processing modes.
- Integrates with error handling, retries, and Dead Letter Topics.
- Flexible method signatures -- payload, full record, or specific headers.

Attribute	Purpose
topics	Topics to subscribe to (property-driven in Lab 31).
groupId	Consumer group id (crm-notifications).
@Header(...)	Access a specific Kafka header, e.g. RECEIVED_KEY.

LAB 31: CustomerEventListener.java

```
@Component
public class CustomerEventListener {

    @KafkaListener(topics = "${crm.kafka.customer-topic}")
    void receive(CustomerEvent event,
        @Header(KafkaHeaders.RECEIVED_KEY) String key) {

        if (!key.equals(event.customerId()))
            throw new InvalidCustomerEventException("key mismatch");

        log.info("customer_event_received id={} customerId={}",
            event.eventId(), event.customerId());

        handler.handle(event);
    }
}
```

KEY TAKEAWAY @KafkaListener declares a method to be a Kafka message listener -- Spring automatically creates the consumer container and invokes the method for each received record.

TRY IT YOURSELF — EXERCISE 2: LISTENER SKETCH

Reinforces: @KafkaListener method signatures, distinct consumer groups on the same topic, and payload typing.

STEP-BY-STEP

STEP	WHAT TO DO
Goal	Sketch two listeners (notifications vs. audit) without compiling code.
Step 1	Method outline: on paper, write @KafkaListener(topics="crm.customer-events.v1", groupId="crm-notifications") void onCustomerEvent(...).
Step 2	Second group: sketch the audit listener with groupId "crm-audit" on the same topic.
Step 3	Payload type: choose String/JsonNode or a typed CustomerEvent DTO -- justify the choice in one line.
Step 4	Correlation: note where correlationId ("lab-request-001") gets logged for support.
Deliverable	Two sketched listeners, distinct group IDs, same topic, plus a typing decision (both groupIds present, same topic, typing + correlation notes written).

SKETCH: TWO @KAFKALISTENER METHODS

```
// Notifications listener (from CustomerEventListener)
@KafkaListener(
    topics = "crm.customer-events.v1",
    groupId = "crm-notifications")
void onCustomerEvent(CustomerEvent event) { ... }

// Exercise 2: sketch the audit listener
@KafkaListener(
    topics = "crm.customer-events.v1",
    groupId = "crm-audit")
void onCustomerEventForAudit(CustomerEvent e) { ... }

// payload type: typed CustomerEvent DTO -- justify in 1
line
// log correlationId ("lab-request-001") for support
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 2 before continuing: [exercises/exercise-02-listener-sketch.md](#).

CONSUMING EVENTS

Consuming events means reading messages from Kafka topics and processing them reliably, while managing offsets and errors correctly.

CONSUMER MODES (DELIVERY SEMANTICS)

Mode	Behavior
At-Least-Once (default)	No message is lost; duplicates may occur -- Lab 31's model.
At-Most-Once	No duplicates, but messages can be lost.
Exactly-Once (EOS)	No duplicates, no loss -- requires idempotent producer + transactions.

BEST PRACTICES

Meaningful groupId

Right offset strategy

Idempotent processing

Monitor consumer lag

Retries + DLT

EXAMPLE (@KafkaListener) -- OrderConsumer.java (general pattern)

```
@Service
public class OrderConsumer {

    @KafkaListener(topics = "orders",
        groupId = "order-service-group",
        containerFactory = "kafkaListenerContainerFactory")
    public void listen(OrderEvent event,
        Acknowledgment ack) {
        // Business logic
        process(event);

        // Manual offset commit only after success
        ack.acknowledge();
    }
}
```

KEY TAKEAWAY Consuming events reliably -- at-least-once or exactly-once -- requires the right offset strategy and idempotent, well-monitored processing logic.

CONSUMER GROUPS IN SPRING

A consumer group is a set of consumers that work together to consume data from topics. Kafka ensures each partition is read by only one consumer in a group.

WHY CONSUMER GROUPS?

- **Scalability** — add more consumer instances to increase throughput.
- **Reliability** — if one consumer fails, others in the group continue.
- **Load Balancing** — partitions are distributed among consumers automatically.
- **Independent Groups** — different groups (e.g. crm-notifications, crm-audit) can read the same topic independently.

Scenario	What Happens
More consumers than partitions	Some consumers stay idle.
A consumer leaves or fails	Its partitions rebalance to the rest of the group.

EXAMPLE: @KAFKALISTENER WITH GROUP ID

```
@KafkaListener(  
    topics = "${crm.kafka.customer-topic}",  
    groupId = "crm-notifications")  
void receive(CustomerEvent event) {  
    // process event once per group  
}
```

BEST PRACTICES

Meaningful groupId

Size group to partitions

Enable retries

Monitor rebalances

KEY TAKEAWAY Consumer groups enable scalable, reliable, and parallel processing. Understanding partition assignment and rebalancing is key to building robust event-driven applications.

TRY IT YOURSELF — EXERCISE 3: FILL SPRING KAFKA TODOS (STARTER)

Reinforces: basic Spring Kafka producer/consumer wiring -- bootstrap-servers, group-id, topic strings, and message keys.

STEP-BY-STEP

STEP	WHAT TO DO
Goal	Fill the TODOs in a tiny Spring Kafka pseudocode snippet -- a STARTER, not a finished solution.
Step 1	Paste the snippet (right) into notes/lab31-todos.md.
Step 2	Fill blanks with: localhost:9092 (or the instructor's bootstrap address), crm-notifications, and crm.customer-events.v1 (used twice).
Step 3	Key reminder: add a comment that the key argument must be CUS-1001 / CUS-1002, never a random UUID.
Step 4	DLT blank: add // TODO Lab 31: route poison messages to crm.customer-events.v1.dlq.
Deliverable	Filled pseudocode with topic/bootstrap/group replaced and key/DLT reminders (all ____ replaced, customer ID key comment present, DLT TODO line present).

STARTER SNIPPET -- FILL EVERY BLANK

```
// application.yml ideas
spring.kafka.bootstrap-servers: ____
spring.kafka.consumer.group-id: ____

@Service
class CustomerEventPublisher {
    private final KafkaTemplate<String, String> template;
    void publishCreated(String customerId, String json) {
        template.send("____", customerId, json); // topic
    }
}

@KafkaListener(topics = "____", groupId = "crm-notifications")
void onEvent(String payload) { /* TODO: parse + idempotent handle */ }

// TODO Lab 31: route poison messages to
//   crm.customer-events.v1.dlq
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 3 before continuing: [exercises/exercise-03-fill-spring-kafka-todos.md](#).

MESSAGE KEYS

A message key is an optional piece of data sent with each record. It determines how the record is stored in Kafka and how ordering is maintained.

WHY USE MESSAGE KEYS?

- **Ordering Guarantee** — records with the same key are always written to, and consumed from, the same partition in order.
- **Data Locality** — related events (same key) stay together in the same partition.
- **Idempotency Support** — keys help implement idempotent writes and deduplication.
- **Better Parallelism** — different keys spread across partitions for higher throughput.

Key Considerations	
null key	Kafka uses round-robin partitioning -- no ordering guarantee.
low-cardinality key	Too few distinct keys causes uneven load (a partition hot-spot).

LAB 31: PUBLISHING WITH A KEY

```
// CUS-1001 is the key for every Amina Khan event
kafkaTemplate.send(
    "crm.customer-events.v1",
    event.customerId(), // key = "CUS-1001"
    event);             // value = CustomerEvent
```

COMMON KEY TYPES

Key Type	Use Case
String	Customer ID, Order ID, User ID (Lab 31: customerId).
UUID	Unique correlation or event IDs.

KEY TAKEAWAY Use message keys to control partitioning, maintain order, and keep related data together. Same key -> same partition -> ordered processing.

OFFSET MANAGEMENT

Offsets represent the position of a consumer in a partition. Managing offsets correctly ensures reliable processing, no data loss, and no duplicate processing.

WHAT IS AN OFFSET?

- A unique, monotonically increasing number identifying a record's position within a partition, starting from 0.
- Offsets are stored by Kafka itself, tracked per consumer group + topic + partition.

Commit Strategy	Description
Auto Commit	Committed automatically at regular intervals (default).
Manual Commit	Application commits after successful processing (recommended).
Commit Sync	Blocks until the offset is successfully committed.
Commit Async	Commits in the background (non-blocking).

AUTO OFFSET RESET (no committed offset exists)

Value	Description
earliest	Start reading from the earliest available offset (Lab 31's setting).
latest	Start reading from the latest offset (new records only).
none	Throw an exception if no committed offset is found.

BEST PRACTICES

Disable auto-commit

Commit after success

Never commit on failure

Monitor consumer lag

Lab 31: enable-auto-commit: false and auto-offset-reset: earliest -- the application controls exactly when progress is recorded.

KEY TAKEAWAY Offsets = progress. Commit them at the right time -- after success, never on failure -- to achieve no data loss and no duplicates.

ERROR HANDLING STRATEGIES

Robust error handling ensures message processing is resilient, failures are isolated, and messages are not lost. Spring Kafka provides multiple mechanisms to handle errors gracefully.

TYPES OF ERRORS

- **DeserializationException** — failed to convert message key/value (bad JSON, wrong trusted package).
- **Non-Retryable Contract Errors** — key mismatch, unsupported event version -- retrying can never fix these.
- **Transient Infrastructure Errors** — broker unavailable, timeout -- often self-resolve.

Strategy	Use Case
Retry	Transient failures (timeouts, temporary unavailability).
Skip	Non-critical messages; bad data that can be ignored.
Dead Letter Topic	Permanent failures, poison messages (Lab 31's approach).
Stop Consumer	Critical, unrecoverable system issues.

LAB 31: KafkaErrorConfig.java

```
@Bean
public DefaultErrorHandler errorHandler(
    KafkaTemplate<Object, Object> template) {
    var recoverer = new DeadLetterPublishingRecoverer(template);
    var handler = new DefaultErrorHandler(
        recoverer, new FixedBackOff(1000, 2));

    handler.addNotRetryableExceptions(
        InvalidCustomerEventException.class,
        UnsupportedEventVersionException.class);
    return handler;
}
```

Bounded retries

Route poison to DLT

Never infinite-retry

KEY TAKEAWAY Choose the right strategy based on the type of error: retry transient failures with backoff, route permanent failures to a Dead Letter Topic, and never let a bad contract retry forever.

DEAD LETTER TOPICS (DLT)

A Dead Letter Topic is a special Kafka topic where messages that cannot be processed successfully are sent, so they are not lost and can be inspected, retried, or fixed later.

WHAT'S PRESERVED IN A DLT RECORD

Header	Example
kafka_dlt-exception-fqcn	InvalidCustomerEventException
kafka_dlt-exception-message	key mismatch
kafka_dlt-original-topic	crm.customer-events.v1
kafka_dlt-original-partition / -offset	1 / 15

DLT NAMING -- LAB 31 REQUIREMENT

Lab 31 requires documenting which naming convention is used: Spring's default .DLT suffix, or Lab 30's .dlq suffix -- an undocumented choice makes .dlq consumers look 'empty' while Spring silently writes elsewhere.

<topic>.DLT (Spring default)

<topic>.dlq (Lab 30)

EXAMPLE: DLT CONSUMER

```
@KafkaListener(topics = "crm.customer-events.v1.DLT",
    groupId = "dlq-consumer-group")
public void listenDlt(ConsumerRecord<String, String> record) {
    log.warn("dlt_message key={} headers={}",
        record.key(), record.headers());
    // Inspect, alert, fix, or trigger reprocessing
}
```

BEST PRACTICES

Monitor DLT size

Alert on growth

Analyze & clean regularly

Avoid infinite reprocess loops

KEY TAKEAWAY A DLT prevents message loss and keeps the main topic flow healthy -- it preserves the original message, headers, and exception info for later inspection, retry, or reprocessing.

TRY IT YOURSELF — EXERCISE 4: ERROR AND DLT NOTES

Reinforces: classifying retryable vs. non-retryable errors and documenting DLT replay ownership.

STEP-BY-STEP

STEP	WHAT TO DO
Goal	Describe when a listener should retry vs. send to the DLT.
Step 1	Retryable example: a transient network blip calling the email API -- retry.
Step 2	Non-retryable example: JSON missing customerId -- send to DLT after limited attempts.
Step 3	Ops note: support replays the DLT after fixing the consumer, using correlationId lab-request-001.
Step 4	No runtime: confirm you will NOT publish to the DLT from the CLI in this pre-lab.
Deliverable	Clear retry-vs-DLT decision notes for Lab 31 (retryable example, non-retryable example, replay/ops sentence).

NOTES: RETRY VS. DLT

```
// notes/lab31-error-dlt.md

Retryable:
  transient network blip calling email API
  -> retry

Non-retryable:
  JSON missing customerId -> DLT after
  limited attempts

Ops note:
  support replays DLT after fixing consumer,
  using correlationId "lab-request-001"

// No runtime: do NOT publish to DLT via CLI here
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 4 before continuing: [exercises/exercise-04-error-dlt-notes.md](#).

RELIABLE EVENT PROCESSING (1 OF 2)

Reliable event processing ensures messages are processed exactly once or at least once without loss or duplication, even in the presence of failures.

DELIVERY SEMANTICS

- **At-Least-Once (default)** — a message may be processed more than once, but will never be lost. Achieved by committing offsets after processing.
- **Exactly-Once (end-to-end)** — a message is processed once and only once. Requires idempotent producer + transactions + atomic offset commit.

KEY RELIABILITY PRINCIPLES

- **Durability** — Kafka replicates data across brokers to prevent loss.
- **Idempotency** — processing the same event twice has the same result.
- **Atomicity** — business logic + offset commit happen as one unit.
- **Consistency** — the system stays valid even after failures.

RELIABLE EVENT PROCESSING FLOW



Commit offsets only AFTER the side effects are successfully completed -- committing first risks silently losing the event if a crash happens mid-processing.

Scenario	Impact	Handling
Consumer crashes	Event may be reprocessed	At-least-once + idempotent logic
Network failure	May cause duplicates	Idempotent producer + consumer

KEY TAKEAWAY Reliable event processing is a combination of Kafka's durability guarantees and application design -- idempotency and atomicity, not Kafka configuration alone.

RELIABLE EVENT PROCESSING (2 OF 2)

Idempotent consumer design is what turns Kafka's at-least-once guarantee into correct, once-only business behavior - built and proven in Lab 31.

LAB 31: ProcessedEventStore.java

```
public class ProcessedEventStore {
    private final Set<UUID> processedIds =
        ConcurrentHashMap.newKeySet();

    public boolean markIfNew(UUID eventId) {
        return processedIds.add(eventId);
    }
}

// in the listener:
if (!store.markIfNew(event.eventId())) {
    log.info("duplicate_event_ignored id={}", event.eventId());
    return;
}
notificationService.notify(event);
```

Mark BEFORE the side effect, not after -- marking after risks a duplicate notification if the process crashes between the side effect and the mark.

COMMON FAILURE SCENARIOS & HANDLING

Scenario	Impact	Strategy
Duplicate events (replay/rebalance)	Duplicate side effects	Unique eventId + ProcessedEventStore
DB failure after processing	Offset not committed	Commit only after successful update
Broker failure	Temporary unavailability	Replication + retries ensure no loss

BEST PRACTICES

- Commit after success only
- Idempotent producers
- Design consumers to be idempotent
- Monitor lag & DLT

Production note: an in-memory Set resets on restart -- production systems use a durable, shared unique-key table (database) instead.

KEY TAKEAWAY Idempotent consumer design (mark-then-notify, never notify-then-mark) is what proves once-only business effects even though Kafka itself only guarantees at-least-once delivery.

TRY IT YOURSELF — EXERCISE 5: IDEMPOTENCY PLAN

Reinforces: designing an idempotency key and processed-event check for duplicate-safe consumers, on paper.

STEP-BY-STEP

STEP	WHAT TO DO
Goal	Define how a consumer ignores a second delivery of Amina's Created event.
Step 1	Why duplicates: list two causes -- producer retry, consumer rebalance/reprocess.
Step 2	Business key: propose an idempotency key, e.g. eventId or customerId+eventType+occurredAt, for CUS-1001.
Step 3	Store idea: one sentence -- check a processed-events table/set before side effects (email).
Step 4	Out of scope: do not implement the table yet -- paper design only.
Deliverable	A short idempotency plan tied to Northstar customer events (two duplicate causes, concrete key proposal, processed-store idea stated).

SKETCH: IDEMPOTENCY NOTES

```
// notes/lab31-idempotency-plan.md

Duplicate cause 1: producer retry
Duplicate cause 2: consumer rebalance / reprocess

Idempotency key:
  customerId + eventType + occurredAt
  (e.g. "CUS-1001")

Store check (before side effect):
  if (!store.markIfNew(eventId)) return;
  // then, and only then: send notification
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 5 before continuing: [exercises/exercise-05-idempotency-plan.md](#).

KAFKA BEST PRACTICES (1 OF 2)

Following best practices helps build reliable, scalable, secure, and maintainable event-driven systems using Apache Kafka.

Category	Practices
1. Topic Design	Use meaningful, business-focused topic names. Keep topics focused on one event type. Choose an appropriate partition count for parallelism. Plan for scalability -- partitions can be increased but never decreased.
2. Producer Practices	Use acks=all for strong delivery guarantees. Enable compression (gzip/snappy/lz4) to reduce network usage. Use meaningful keys with good cardinality. Set timeouts (linger.ms, batch.size) properly. Monitor producer metrics.
3. Consumer Practices	Use consumer groups for parallel processing and availability. Commit offsets after successful processing (manual preferred). Handle failures gracefully with retries and a DLT. Tune fetch settings (max.poll.records) for the workload.
4. Message Design	Use message keys to control ordering and partitioning. Use structured formats (JSON, Avro/Protobuf with a schema registry). Include metadata (timestamps, schema version, correlationId). Make messages immutable -- publish new events for changes.

KEY TAKEAWAY Design topics, producers, consumers, and message contracts deliberately -- these four foundations determine everything about how reliable and scalable your Kafka integration will be.

KAFKA BEST PRACTICES (2 OF 2)

Reliability, security, performance, operations, and development practices for production-ready Kafka integrations.

Category	Practices
5. Reliability & Fault Tolerance	Use adequate replication (RF >= 3 for production). Enable idempotent producers (enable.idempotence=true). Use transactions for exactly-once when truly required. Send failed records to a DLT -- never lose bad data. Use exponential backoff with bounded retries.
6. Security	Enable authentication (SASL/OAUTHBEARER) for clients and brokers. Enable authorization (ACLs) to control access to topics and groups. Encrypt data with TLS/SSL in transit and at rest.
7. Performance & Scalability	Right-size partitions -- more partitions means more parallelism, but avoid too many small ones. Use fast disks (SSD), sufficient CPU/memory/network. Monitor broker metrics and scale horizontally when needed.
8. Operations & Monitoring	Monitor broker health, consumer lag, throughput, error rates, and disk usage. Set up alerts for high lag, disk usage, and under-replicated partitions. Plan capacity for data growth. Perform regular maintenance and version updates.
9. Development Practices	Externalize configuration using Spring Profiles and environment variables. Test thoroughly with realistic data and failure scenarios. Use Docker Compose / Testcontainers for local development and testing.

KEY TAKEAWAY Design for reliability, build for scale, secure by default, and monitor continuously -- these are the operating principles that keep Kafka production-ready over time.

TRY IT YOURSELF — EXERCISE 6: LAB 31 READINESS

Reinforces: self-checking Lab 30 dependencies, tooling, and expected evidence before starting Lab 31.

STEP-BY-STEP

STEP	WHAT TO DO
Goal	Confirm Lab 30 topic names are ready inputs for Spring Kafka wiring.
Step 1	Dependency: write that Lab 31 assumes <code>crm.customer-events.v1</code> (+ DLQ) naming from Lab 30.
Step 2	Maven habit: note you will use JDK 21 + Maven with the <code>spring-kafka</code> dependency in the starter (not yet).
Step 3	Evidence preview: list evidence expected later -- a log line for Amina's event, listener metrics/notes.
Step 4	Pass mark: pass if Exercises 2-4 notes exist; otherwise revisit the TODOs.
Deliverable	A readiness note linking Lab 30 topics to the Spring Kafka lab goals (topic dependency stated, JDK 21/Maven note, pass/fail self-mark).

NOTES: LAB 31 READINESS

```
// notes/lab31-readiness.md

Dependency: crm.customer-events.v1 (+ DLQ)
           naming comes from Lab 30

Tooling: JDK 21 + Maven; spring-kafka
        dependency added in the lab starter

Evidence expected: log line for Amina's event,
                  listener metrics/notes

Pass mark: Exercises 2-4 notes exist,
           else revisit the TODOs
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 6 before continuing: `exercises/exercise-06-lab31-readiness.md`.

LAB OVERVIEW -- SPRING BOOT INTEGRATION WITH KAFKA

Lab 31: Spring Boot Integration with Kafka -- Northstar CRM Listeners

BUSINESS SCENARIO

- Lab 30 proved the broker. Leadership now requires the CRM service itself to emit and consume versioned customer events -- without losing notifications on poison messages or double-processing replays.
- When an agent creates Amina (CUS-1001) or activates Ravi (CUS-1002), notification and audit processes must learn asynchronously; duplicate delivery must not send duplicate SMS/email side effects.
- Leadership freezes the rule: publish keyed CustomerEvent v1 after successful writes; consume idempotently; recover poison messages to a DLT after bounded retry.

LEARNING OBJECTIVES

- Add Spring Kafka dependencies to the CRM service.
- Configure JSON producers/consumers with externalized topic names.
- Publish typed customer events with KafkaTemplate.
- Write @KafkaListener handlers validating key, version, and metadata.
- Configure bounded retries and dead-letter publishing.
- Write idempotent event-processing logic; test with Embedded Kafka.

WHAT YOU BUILD

- Copy lab29-crm to lab31-crm; add spring-kafka (+test); define immutable CustomerEvent; publish with KafkaTemplate keyed by customerId; write @KafkaListener validating key<->payload; add ProcessedEventStore idempotency; configure DefaultErrorHandler + DeadLetterPublishingRecoverer with non-retryable contract errors; write an integration test that awaits a handled event.

KEY TAKEAWAY Fixtures CUS-1001 (Amina Khan, ACTIVE) and CUS-1002 (Ravi Singh, PROSPECT) anchor every example; topic crm.customer-events.v1, group crm-notifications, and correlation ID lab-request-001 appear throughout.

LAB TASKS AND DELIVERABLES

Northstar CRM Listeners -- implementation steps, deliverables, and the evaluation rubric.

#	Implementation Step
1	Branch CRM to lab31-crm; add spring-kafka (+ spring-kafka-test).
2	Configure JSON messaging: bootstrap servers, group id, trusted packages, topic name -- all externalized.
3	Define CustomerEvent record (eventId, eventType, eventVersion, customerId, correlationId, data) with null/version guards.
4	Publish with KafkaTemplate after successful writes; log publish success/failure with partition/offset.
5	Write a typed @KafkaListener validating key == customerId; log received event with correlationId.
6	Make processing idempotent: ProcessedEventStore.markIfNew(eventId) before notifying.
7	Configure DefaultErrorHandler + DeadLetterPublishingRecoverer with FixedBackOff and non-retryable exceptions.
8	Integration test with EmbeddedKafka/Testcontainers awaiting a specific eventId; document DLT naming + runbook.

EXPECTED DELIVERABLES

- Spring Kafka publisher + listener for CRM events.
- Idempotent processing evidence.
- Retry + DLT configuration with proof.
- Integration test output (EmbeddedKafka/Testcontainers).
- Successful Amina/Ravi path evidence.
- Controlled poison-message / duplicate evidence.
- Runbook + DLT naming notes; no secrets committed.

RUBRIC (100 MARKS)

- Environment 10 · Core Impl 30 · Integration/Config 15 · Failure Handling 15 · Verification 10 · Security/Prod Awareness 10 · Docs 10

KEY TAKEAWAY Listener without idempotency -> incomplete. Infinite retry on bad contracts -> honor issue. Sleep-only tests -> quality marks lost (per the guide's own explicit notes).

MODULE 31 SUMMARY

In this module, we learned how to integrate Apache Kafka with Spring Boot to build reliable, scalable, event-driven microservices.

WHAT WE COVERED



Consumer Groups

Enable load balancing and scalability across consumers.



Message Keys

Ensure ordering and control partitioning strategy.



Offset Management

Auto-commit, manual commit, and offset reset strategies.



Dead Letter Queues

Capture and isolate failed messages for retry or analysis.



Reliable Processing

Achieve at-least-once and exactly-once semantics.



Kafka Best Practices

Design topics, producers, and consumers for production.

MODULE FLOW RECAP

1

KafkaTemplate Publish

2

@KafkaListener Consume

3

Idempotent Processing

4

Retry + DLT

5

Lab: Northstar CRM

KEY TAKEAWAY Effective Kafka integration is essential for building enterprise-grade, event-driven APIs. It ensures scalability, reliability, and resilient, production-ready event processing.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of Spring Kafka components, consumer groups, message keys, and offset behavior.

1. Which Kafka component stores messages durably?

- A. Zookeeper
- B. Broker**
- C. Producer
- D. Consumer

3. What does a message key help achieve?

- A. Reduce message size
- B. Ensure ordering and partitioning**
- C. Increase throughput
- D. Enable compression

2. What is the primary purpose of a Consumer Group?

- A. To increase message retention
- B. To enable parallel consumption**
- C. To replicate messages
- D. To encrypt messages

4. What happens when a consumer tries to read a topic with no committed offset available?

- A. It always reads from the latest offset
- B. It always throws an error
- C. It starts from earliest or latest, based on auto-offset-reset**
- D. It always replays from the beginning

KEY TAKEAWAY Brokers store messages durably; consumer groups enable parallel processing; keys guarantee ordering; auto-offset-reset controls behavior when no committed offset exists.

KNOWLEDGE CHECK (2 OF 2)

Four more questions on delivery guarantees, Dead Letter Queues, producer acknowledgments, and testing tools.

5. Which delivery guarantee ensures that messages are processed exactly once?

- A. At-most-once
- B. At-least-once
- C. Exactly-once**
- D. Best-effort once

7. Which producer setting ensures a message is written to all in-sync replicas?

- A. acks=0
- B. acks=1
- C. acks=all**
- D. acks=-1 (a distinct setting from acks=all)

6. What is the purpose of a Dead Letter Queue (DLQ)?

- A. To increase topic partitions
- B. To store successfully processed messages
- C. To capture failed messages for retry or analysis**
- D. To speed up consumers

8. Which tool is used in this module's lab to test Kafka-integrated REST APIs?

- A. Postman**
- B. Maven
- C. JUnit
- D. Docker

KEY TAKEAWAY Exactly-once needs idempotent producers + transactions; a DLQ captures failed messages for later analysis; acks=all waits for every in-sync replica; Postman exercises the REST layer around Kafka.

MODULE 31 COMPLETE

Kafka Integration with Spring Boot

You can now publish events with `KafkaTemplate`, consume them reliably with `@KafkaListener`, build idempotent processing, and recover failures with retries and Dead Letter Topics.